Prop (Properties of equivalence classes)

① $a \in C(a) \quad \forall a \in A$.

② If $a, b \in A$ satisfy $a \sim b$, then $C(a) = C(b)$.

③ If $a, b \in A$ do not satisfy $a \sim b$, then $C(a) \cap C(b) = \emptyset$.

④ $A = \bigcup_{a \in A} C(a)$.

Pf: ① and ②: Last time.

③ Suppose not: $\exists\, a, b \in A$ s.t. $a \not\sim b$, but $C(a) \cap C(b) \neq \emptyset$. Then $\exists\, c \in C(a) \cap C(b)$, and

$$\left.\begin{array}{lll} c \in C(a) & \Rightarrow & c \sim a \overset{\text{symmetry}}{\Rightarrow} a \sim c \\ c \in C(b) & \Rightarrow & c \sim b \end{array}\right\} \underset{\text{transitivity}}{\Rightarrow} a \sim b.$$

This contradicts that $a \not\sim b$.

④ By definition $C(a) \subseteq A \;\forall a \in A$, and so

$$\bigcup_{a \in A} C(a) \subseteq A$$

On the other hand,

$$A = \bigcup_{a \in A} \{a\} \overset{①}{\subseteq} \bigcup_{a \in A} C(a)$$

Together, we conclude $A = \bigcup_{a \in A} C(a)$. $\square$

So far: $\mathbb{N} \rightsquigarrow \mathbb{Z} \rightsquigarrow \mathbb{Q}$ ($\rightsquigarrow \mathbb{R}$, next week)

Definition:	✓	✓	✓
Countable?	✓	✓	✓
$+, \cdot$:	✓	✓	today

<u>Recall</u>: • $(a,b), (c,d) \in \mathbb{Z} \times \mathbb{N}$, $(a,b) \sim (c,d) \iff ad = bc$

• $C((a,b)) = \{ (c,d) \in \mathbb{Z} \times \mathbb{N} : (c,d) \sim (a,b) \}$

The set of <u>rational numbers</u> is

$$\mathbb{Q} = \{ C((a,b)) : a \in \mathbb{Z}, b \in \mathbb{N} \}$$
$$= \{ \tfrac{a}{b} : a \in \mathbb{Z}, b \in \mathbb{N} \}$$

For $(a,b) \in \mathbb{Z} \times \mathbb{N}$, we denote its equivalence class $C((a,b))$ by $\frac{a}{b}$. We also set $\frac{a}{-b} = \frac{-a}{b}$ for $b \in \mathbb{N}$.

• E.g., what is $\frac{1}{2}$? $\frac{1}{2}$ is the equivalence class $\{(1,2), (2,4), (3,6), \dots\}$ of all fractions equivalent to $\frac{1}{2}$.

We define

① $\frac{a}{b} + \frac{c}{d} = \frac{ad+bc}{bd}$ (i.e. $(a,b) + (c,d) = (ad+bc, bd)$)

② $\frac{a}{b} \cdot \frac{c}{d} = \frac{ac}{bd}$ (i.e. $(a,b) \cdot (c,d) = (ac, bd)$)

• These definitions only use $+, \cdot$ on $\mathbb{Z}$, and so they only need the Peano axioms

We have to check that these are well-defined:
If $(a,b) \sim (a',b')$ and $(c,d) \sim (c',d')$, then

① $(ad+bc, bd) \sim (a'd'+b'c', b'd')$

② $(ac, bd) \sim (a'c', b'd')$

<u>Pf of ①</u>: Want: $(ad+bc)b'd' = bd(a'd'+b'c')$.

We know

$$(a,b) \sim (a',b') \Rightarrow ab' = ba' \overset{\cdot dd'}{\Rightarrow} adb'd' = a'd'bd$$
$$(c,d) \sim (c',d') \Rightarrow cd' = dc' \underset{\cdot bb'}{\Rightarrow} bcb'd' = bdb'c'$$

Add these together:

$$\underbrace{adb'd' + bcb'd'}_{(ad+bc)b'd'} = \underbrace{ad'bd + bdb'c'}_{bd(ad'+b'c')}$$

$\square$

<u>Exc</u> Prove ②.

FIELDS (Rudin Ch. 1)

<u>Def</u> $(F, +, \cdot)$ is a <u>field</u> if F is a set with at least two elements, and the operations $+, \cdot$ satisfy:

(A1) Closure: if $a,b \in F$ then $a+b \in F$

(A2) Commutativity: if $a,b \in F$ then $a+b = b+a$

(A3) Associativity: if $a,b,c \in F$ then $(a+b)+c = a+(b+c)$

(A4) Identity: $\exists\, 0 \in F$ s.t. $a+0 = 0+a = a \;\; \forall a \in F$

(A5) Inverse: $\forall a \in F \;\; \exists\, (-a) \in F$ s.t. $a+(-a) = -a+a = 0$

(M1) Closure: if $a,b \in F$ then $a \cdot b \in F$

(M2) Commutativity: if $a,b \in F$ then $a \cdot b = b \cdot a$

(M3) Associativity: if $a,b,c \in F$ then $(a \cdot b) \cdot c = a \cdot (b \cdot c)$

(M4) Identity: $\exists\, 1 \in F$ s.t. $a \cdot 1 = 1 \cdot a = a \;\; \forall a \in F$

(M5) Inverse: $\forall a \in F \setminus \{0\} \;\; \exists\, a^{-1} \in F$ s.t. $a \cdot a^{-1} = a^{-1} \cdot a = 1$

(D) Distributivity: if $a,b,c \in F$ then $(a+b) \cdot c = a \cdot c + b \cdot c$

<u>Ex</u> $(\mathbb{N}, +, \cdot)$ is not a field: (A4) fails

<u>Ex</u> $(\mathbb{Z}, +, \cdot)$ is not a field: (M5) fails

<u>Ex</u> $(\mathbb{Q}, +, \cdot)$ is a field!

- Additive identity is $0 \overset{\text{def}}{=} \frac{0}{1}$ = equivalence class of $(0,1)$
- Multiplicative identity is $1 \overset{\text{def}}{=} \frac{1}{1}$ = equivalence class of $(1,1)$
- Additive inverse of $\frac{a}{b}$ is $-\frac{a}{b} \overset{\text{def}}{=} \frac{-a}{b}$
- Multiplicative inverse of $\frac{a}{b} \neq 0$ is $\left(\frac{a}{b}\right)^{-1} \overset{\text{def}}{=} \begin{cases} \frac{b}{a}, & a>0 \\ \frac{-b}{|a|}, & a<0 \end{cases}$

Exc: Properties (A1) – (D) hold.

<u>Prop</u> Let $(F, +, \cdot)$ be a field. Then:

① The additive and multiplicative identities are unique.

② Additive and multiplicative inverses are unique.

③ If $a, b, c \in F$ satisfy $a+b = a+c$, then $b=c$.
If $a, b, c \in F$ satisfy $a \cdot b = a \cdot c$ and $a \neq 0$, then $b=c$.

④ $a \cdot 0 = 0 \cdot a = 0 \quad \forall a \in F$

⑤ $(-a) \cdot b = a \cdot (-b) = -(a \cdot b) \quad \forall a, b \in F$

⑥ $(-a) \cdot (-b) = a \cdot b \quad \forall a, b \in F$

⑦ If $a \cdot b = 0$ then $a = 0$ or $b = 0$.